3.6.4 Homeostasis is the maintenance of a stable internal environment (A-level only)

3.6.4.1 Principles of homeostasis and negative feedback (A-level only)

Content	Opportunities for skills development
Homeostasis in mammals involves physiological control systems that maintain the internal environment within restricted limits. The importance of maintaining a stable core temperature and stable blood pH in relation to enzyme activity. The importance of maintaining a stable blood glucose concentration in terms of availability of respiratory substrate and of the water potential of blood. Negative feedback restores systems to their original level. The possession of separate mechanisms involving negative feedback controls departures in different directions from the original state, giving a greater degree of control. **Students should be able to** interpret information relating to examples of negative and positive feedback.	

3.6.4.2 Control of blood glucose concentration (A-level only)

<table>
<tr><th>Content</th><th>Opportunities for skills development</th></tr>
<tr><td>
The factors that influence blood glucose concentration.
The role of the liver in glycogenesis, glycogenolysis and gluconeogenesis.
The action of insulin by:

attaching to receptors on the surfaces of target cells
controlling the uptake of glucose by regulating the inclusion of channel proteins in the surface membranes of target cells
activating enzymes involved in the conversion of glucose to glycogen.

The action of glucagon by:

attaching to receptors on the surfaces of target cells
activating enzymes involved in the conversion of glycogen to glucose
activating enzymes involved in the conversion of glycerol and amino acids into glucose.

The role of adrenaline by:

attaching to receptors on the surfaces of target cells
activating enzymes involved in the conversion of glycogen to glucose.

The second messenger model of adrenaline and glucagon action, involving adenylate cyclase, cyclic AMP (cAMP) and protein kinase.
The causes of types I and II diabetes and their control by insulin and/or manipulation of the diet.
Students should be able to evaluate the positions of health advisers and the food industry in relation to the increased incidence of type II diabetes.
</td><td></td></tr>
<tr><td>Required practical 11: Production of a dilution series of a glucose solution and use of colorimetric techniques to produce a calibration curve with which to identify the concentration of glucose in an unknown ‘urine’ sample.</td><td>AT b and c</td></tr>
</table>

3.6.4.3 Control of blood water potential (A-level only)

Content	Opportunities for skills development
Osmoregulation as control of the water potential of the blood. The roles of the hypothalamus, posterior pituitary and antidiuretic hormone (ADH) in osmoregulation. The structure of the nephron and its role in: • the formation of glomerular filtrate • reabsorption of glucose and water by the proximal convoluted tubule • maintaining a gradient of sodium ions in the medulla by the loop of Henle • reabsorption of water by the distal convoluted tubule and collecting ducts.	